GRAYLINX

AI Platform — Feature Review Pack

Every feature on the platform, in one place, for product and SME review. Companion to the full Product & Architecture Document v4.

1. How to Use This Pack

This is a working document, not a reference document. Every feature and every open question carries an ID. Mark each row, then record what was agreed in the decision log at the end. The full detail behind any row is in the main document — this pack exists so that a review meeting can cover the whole platform without reading seventy-eight pages.

Mark	Meaning	What happens next
A	Agreed as written	No action — goes into the build backlog as specified
C	Change requested	Note the change in the decision log; the main document is updated
Q	Question / needs data	Owner and due date recorded; blocks the feature until answered
D	Defer	Moves to a later phase; recorded with the reason
X	Out of scope	Removed from the feature register with a reason

Engine column: SW = plain deterministic software · R = deterministic rule · ML = equipment machine-learning model · LLM = language model / agentic layer. Anything marked R or SW must produce the same answer every time from the same inputs, and is testable line by line.

2. The Platform on One Page

ASK → UNDERSTAND → FIND DATA → EXPLAIN → RECOMMEND → CREATE / ACT → VERIFY → REPORT → LEARN

Layer	What sits here	Engine	Who owns it
Data and digital twin	BMS acquisition, tag normalisation, hierarchy, quality flags	SW	Data / IoT
Equipment ML	Six normal-operation models per chiller; residuals and validity flags	ML	Reliability + Data
FDD and reliability	Operating gates, persistence, isolation path, fault class, confidence	R	Reliability + SME
Knowledge	SOPs, manuals, WO history, verified cases — retrieved with sources	LLM + retrieval	Maintenance
AI Copilot	Understanding, explanation, planning, drafting, summarising	LLM	Product + AI
Control Plane	Identity, scope, risk, approval, tool gateway, audit	SW	Platform + Security
Execution	Work orders, planning, inventory, notifications, reports	SW	Maintenance + Platform

Priority	Pillar	One-line promise	Feature count in this pack
P0	AI Copilot	One assistant that understands, explains, prepares work and proves the result.	14
P0	Reports	Ask a question, get the answer with the numbers behind it.	10
P0	Work Orders	Work that arrives carrying its own justification and cannot close unproven.	12
P0	Reliability / FDD	Find the fault, name one cause, or say honestly that we cannot.	14
P0	Control Plane	Nothing acts without identity, scope, approval and an audit record.	8
P1	Asset, Knowledge, Planning, Inventory, Alerts, Verification, Roles, Safety	Everything the loop needs to close.	32

3. Feature Register

Eighty-five features across thirteen domains. This is the complete scope of the new AI layer — if something is not on this list, it is not being built.

3.1 AI Copilot

ID	Feature	What it does	Primary user	Engine	Pri	Mark
C1	Natural-language ask across the platform	One entry point to reports, assets, faults, work orders and documents.	Everyone	LLM + SW	P0	
C2	Context resolution	Understands “this” and “it” from the screen and the conversation.	Everyone	LLM + SW	P0	
C3	Intent routing	Chooses the right skill and tools for the request.	Everyone	LLM	P0	
C4	Evidence pack assembly	Collects the readings, versions and records behind every answer.	Everyone	SW	P0	
C5	Grounded fault explanation	Explains the FDD rule path in plain English, without inventing it.	Reliability	LLM	P0	
C6	Ranked recommendation	Proposes next actions with reason, confidence and alternatives.	Maintenance	LLM + R	P0	
C7	Honest uncertainty	Returns NO_DIAGNOSIS when gates fail or signals conflict.	Everyone	R	P0	
C8	Draft before write	Shows the complete action before anything is saved.	Everyone	SW	P0	
C9	Approval request routing	Prepares the request and names the approver; never self-approves.	Supervisor	SW	P0	
C10	Multi-step task memory	Carries a job from investigation to work order to verification.	Everyone	SW	P0	
C11	Role-scoped answers	Same assistant, different scope and depth per role.	Everyone	SW	P0	
C12	Source-cited knowledge answers	Quotes the SOP or manual and shows where it came from.	Technician	LLM	P0	
C13	Watches	“Tell me if this gets worse” — a monitored condition with a notification.	Plant Manager	R	P1	
C14	Executive briefing	Condenses the week into a short business summary with drill-down.	Executive	LLM	P1	

3.2 Reports

ID	Feature	What it does	Primary user	Engine	Pri	Mark
R1	Ask a question of a report	Conversational follow-up on any report.	All	LLM	P0	
R2	Create report from natural language	Builds a report from a spoken or typed request.	Manager	LLM + SW	P0	
R3	Explain a KPI change	Traces a movement to the equipment, faults and work behind it.	Manager	LLM + SW	P0	
R4	Compare sites, assets or periods	Aligns two scopes on identical metric definitions.	Regional Head	SW	P0	
R5	Drill down to source records	Every number opens to the records that produced it.	Analyst	SW	P0	
R6	AI narrative	Writes the commentary over verified metrics only.	Executive	LLM	P1	
R7	Schedule and deliver	Recurring role-aware delivery.	Manager	SW	P1	
R8	Export with lineage	Export carries metric definitions and source references.	Analyst	SW	P1	
R9	Customer-scoped report	Customers see only their own sites and assets.	Customer	SW	P0	
R10	KPI reconciliation	Reported numbers are checked against the source calculation.	Finance	SW	P0	

3.3 Work Orders

ID	Feature	What it does	Primary user	Engine	Pri	Mark
W1	Create work order from chat	From a sentence, with the evidence attached.	Maintenance	LLM + SW	P0	
W2	Create work order from a fault	Direct from an FDD finding, pre-populated.	Reliability	R + SW	P0	
W3	Evidence auto-attached	Residuals, gates, trends and similar cases travel with the job.	Technician	SW	P0	
W4	Priority calculation	Deterministic formula over criticality, risk, SLA and production impact.	Planner	R	P0	

ID	Feature	What it does	Primary user	Engine	Pri	Mark
W5	Maintenance window planning	Finds a production-compatible slot.	Planner	SW	P1	
W6	Technician assignment	Matches skills, certifications and availability.	Supervisor	SW	P1	
W7	Parts check and reservation	Stock, alternates, delivery impact.	Stores	SW	P1	
W8	Findings capture	Structured technician findings, readings and photographs.	Technician	SW	P0	
W9	Verification gate before close	A work order cannot close unproven.	Supervisor	R	P0	
W10	Reopen on failed repair	Previous findings preserved for the next technician.	Supervisor	SW	P0	
W11	Closure summary	What was found, what was done, what it proved.	Manager	LLM	P1	
W12	SLA escalation	Routes when work approaches or breaches its SLA.	Manager	R	P1	

3.4 Asset Intelligence

ID	Feature	What it does	Primary user	Engine	Pri	Mark
A1	One equipment story	Health, status, faults, history, documents and risks in one view.	All	SW + LLM	P1	
A2	Health score with reasons	Not just a number — the contributing factors.	Reliability	R + ML	P1	
A3	Dependency and served load	What this machine feeds and what feeds it.	Operations	SW	P1	
A4	Like-for-like comparison	Compares similar machines under similar conditions only.	Reliability	SW	P1	
A5	Repeat-failure / bad-actor detection	Surfaces the assets consuming disproportionate effort.	Reliability	R	P1	

3.5 Reliability and Fault Detection — primary SME review area

ID	Feature	What it does	Primary user	Engine	Pri	Mark
F1	Six normal-operation models per chiller	DP, SP, DT, Power, Compressor Amps, Condenser Leaving.	Reliability	ML	P0	
F2	Residual computation	Actual minus predicted, with a validity flag per residual.	Reliability	ML	P0	
F3	Operating gates	Running steady, load above floor, flows valid, no setpoint change.	Reliability	R	P0	
F4	Persistence and volatility test	Separates a fault from a passing disturbance.	Reliability	R	P0	
F5	Deterministic isolation path	Power → head → water side vs refrigerant side → single class.	Reliability	R	P0	
F6	Sensor bias / model mismatch detection	Rules out instrumentation before dispatching a crew.	Reliability	R	P0	
F7	Honest ambiguity labels	High head — ambiguous; undercharge and restriction kept combined.	Reliability	R	P0	
F8	NO_DIAGNOSIS on failed gates	No softened guess when the inputs are not trustworthy.	Reliability	R	P0	
F9	Similar verified case matching	Finds past cases with the same signature and a proven fix.	Maintenance	SW	P1	
F10	Model health and drift monitoring	Detects a model going quietly stale.	Data / Reliability	ML + R	P0	
F11	Model quarantine	A failed model stops being used, not merely flagged.	Data / Reliability	SW	P0	
F12	Re-baseline after verified major work	A repaired machine may have a legitimately new normal.	Reliability	ML	P1	
F13	Cooling tower assessment	Wet bulb, approach and tower running hours.	Reliability	ML + R	P1	
F14	Chilled-water ΔT health check	Flags sustained low ΔT against the design band.	Reliability	R	P0	

3.6 Knowledge

ID	Feature	What it does	Primary user	Engine	Pri	Mark
K1	SOP search	Approved procedures, retrieved with the source shown.	Technician	LLM	P1	
K2	Manufacturer manual search	What the OEM actually says about this model.	Technician	LLM	P1	
K3	Work order history search	"How was this fixed last time?"	Maintenance	LLM	P1	

ID	Feature	What it does	Primary user	Engine	Pri	Mark
K4	Verified case search	Only fixes that were proven to work.	Reliability	LLM	P1	
K5	Source-visible answers	Every important answer names its document and version.	All	SW	P0	
K6	Explain a technical document simply	Turns dense OEM text into plain English.	Technician	LLM	P2	

3.7 Planning and Inventory

ID	Feature	What it does	Primary user	Engine	Pri	Mark
P1	Prioritised work queue	Ranked by risk, criticality, SLA and production impact.	Planner	R	P1	
P2	Maintenance window finder	Slots that production can actually give.	Planner	SW	P1	
P3	Certification-aware matching	Only people qualified for the task.	Supervisor	SW	P1	
P4	Workload balance	Avoids overloading one crew.	Supervisor	SW	P2	
P5	Plan feasibility answer	“Can we actually do this tomorrow?” in one turn.	Planner	SW + LLM	P1	
I1	Stock check	Availability and location.	Stores	SW	P1	
I2	Reservation against a work order	Parts held for the job.	Stores	SW	P1	
I3	Alternates	Acceptable substitutes when the primary part is out.	Stores	SW	P2	
I4	Delivery impact on schedule	What the lead time does to the plan.	Planner	SW	P2	

3.8 Alerts and Verification

ID	Feature	What it does	Primary user	Engine	Pri	Mark
L1	Duplicate grouping	Forty alerts become one problem.	Operations	R	P1	
L2	Alert explanation	Why this fired, in plain English.	Operations	LLM	P1	
L3	Prioritisation	Ranked, not listed.	Operations	R	P1	
L4	Routing	To the person who can act on it.	Operations	SW	P1	
L5	Escalation	When it is not picked up or crosses a threshold.	Manager	R	P1	
L6	Noise measurement	Alert volume tracked as a product metric.	Product	SW	P2	
V1	Post-work residual comparison	Are the residuals back inside the band?	Reliability	ML + R	P0	
V2	Valid comparison window	Comparison only under comparable operating conditions.	Reliability	R	P0	
V3	Persistence across load range	Improvement must hold, not appear once.	Reliability	R	P0	
V4	Fault-clear check	Does the FDD fault clear under valid conditions?	Reliability	R	P0	
V5	Business outcome check	Did the expected energy or downtime improvement appear?	Manager	SW	P1	
V6	PASS / FAIL / UNKNOWN	Three outcomes — UNKNOWN is a real and common answer.	Supervisor	R	P0	
V7	Verified-case memory	Signature, fix and proven outcome written back for reuse.	Reliability	SW	P1	

3.9 Roles, Safety and Control Plane

ID	Feature	What it does	Primary user	Engine	Pri	Mark
U1	Customer view	Only their own sites, assets and reports.	Customer	SW	P0	
U2	Executive brief	The answer, not twenty screens.	Executive	LLM	P1	
U3	Technician job pack	Job context, safety, SOP, history, parts, findings.	Technician	LLM + SW	P0	
U4	Planner queue view	What can be done, when, by whom.	Planner	SW	P1	
U5	Permission explanation	“Why can’t this user see this report?”	Administrator	SW	P2	

ID	Feature	What it does	Primary user	Engine	Pri	Mark
S1	Safety-critical action block	The AI stops; it does not weigh the risk itself.	EHS	SW	P0	
S2	Permit and isolation gate	Permit, LOTO and isolation checks before work.	EHS	SW	P0	
S3	Qualification check	Only qualified people are proposed for restricted work.	Supervisor	SW	P0	
S4	Safety answers from the SOP	Never answered from model memory.	Technician	LLM	P0	
S5	EHS escalation	Routed per the escalation matrix.	EHS	R	P0	
G1	Identity and scope per turn	Scope recomputed every turn, never inherited.	Platform	SW	P0	
G2	Risk classification	Low / medium / high / safety-critical / system-critical.	Platform	R	P0	
G3	Approval engine	Who must approve what, under which conditions.	Platform	SW	P0	
G4	Tool gateway	Schema validation, credentials, safe retry, side-effect control.	Platform	SW	P0	
G5	Idempotency	A retry can never create a second work order.	Platform	SW	P0	
G6	Audit trail	Every material action and decision, permanently.	Audit	SW	P0	
G7	Break-glass	Emergency elevation that is temporary, recorded and expiring.	Security	SW	P1	
G8	Policy versioning and simulation	Test a rule change before it goes live.	Governance	SW	P1	

4. The Copilot at a Glance

Mode	Trigger	Writes?	Mode	Trigger	Writes?
ASK	A direct question of fact	No	APPROVE	A named human must agree	No
EXPLAIN	Why is this happening?	No	EXECUTE	Confirmed action	Yes
INVESTIGATE	Multi-step enquiry	No	VERIFY	Did the repair work?	Result
COMPARE	Two scopes or periods	No	ESCALATE	Safety or rule-driven routing	Yes
SEARCH	Documents and history	No	MONITOR	Tell me if it gets worse	Config
RECOMMEND	What should we do?	No	SUMMARIZE	Brief me	No
REPORT	Generate a report	Artifact	DRAFT	Prepare an action	No

Every turn ends in exactly one of six states

ANSWERED	PARTIAL	NO_DIAGNOSIS	NEEDS_APPROVAL	BLOCKED	FAILED
Evidence sufficient and grounded.	Part answerable; the rest stated as missing.	Gates failed or signals conflict — with the data-quality action.	Permitted, but a named human must agree.	Scope, permission or safety prevents it.	A tool failed — with what was and was not done.

THE ONE LINE THAT MATTERS

The language model never names a fault and never grants a permission. The FDD rules name the fault; the Control Plane grants the permission. The Copilot explains both.

5. FDD Engine — For SME Review

5.1 The six models and their inputs

Model	Predicts	Inputs	Residual	Mark
DP	Discharge pressure	Suction pressure; cooling load; condenser water entry temp; condenser flow; lead/lag	rDP	
SP	Suction pressure	Evaporator leaving; evaporator entry; evaporator flow / DPT; lead/lag	rSP	
DT	Discharge temperature	Suction pressure; discharge pressure; cooling load; condenser water entry temp; condenser flow; lead/lag	rDT	

Model	Predicts	Inputs	Residual	Mark
Power	Chiller amps / power	Evaporator leaving temp; chiller load; condenser water entry temp; condenser flow	rPwr	
Comp Amps	Compressor amps	Suction pressure; discharge pressure; chiller load; lead/lag	rAmp	
Cond Leaving	Condenser water leaving temp	Chiller load; condenser water entry temp; condenser flow; lead/lag	rCWL	

5.2 Gates — nothing is diagnosed until all pass

Gate	Proposed setting	Confirm with SME
Running steady	No start, stop or stage transient in the window	Settling time after a stage change?
Load above minimum	Above the model's valid load floor	What percentage load is the floor?
Flows valid	Evaporator and condenser flow signals healthy	Is condenser flow actually measured on site?
No setpoint change	No leaving-water setpoint move in the window	How long after a setpoint move before we trust residuals?
Persistence	Pattern held 20–30 minutes	Same window for every fault class, or per class?

5.3 Fault fingerprints — the isolation path

Fault class	rPwr	rDP	rCWL	rSP	rAmp	Separator	Mark
Condenser water-side — low flow	High	High	High	—	—	Spiky / intermittent	
Condenser water-side — fouling	High	High	High	—	—	Steady drift over weeks	
Refrigerant-side high head	High	High	Normal	—	—	Water loop looks normal	
High head — ambiguous	High	High	Noisy	—	—	Needs operator confirmation	
Compressor inefficiency	High	Normal	—	Normal	High	Current high for the pressures	
Starved evaporator	High	Normal	—	Low	—	Suction below expected	
Expansion valve hunting	High	Normal	—	Oscillating	—	rSP and rDT swing repeatedly	
Persistent starvation	High	Normal	—	Low, steady	—	Undercharge and restriction kept combined	
DP sensor bias / model mismatch	Normal	High	Normal	—	Normal	Everything else normal	
DT sensor bias	Normal	Normal	—	—	Normal	Only rDT abnormal	
No diagnosis	any	any	any	any	any	A gate failed or signals conflict	

rDT is a support signal only. It is not used to separate the high-head causes.

5.4 What happens when a signal is missing

Missing signal	Models lost	Diagnostic capability lost	Platform behaviour
Condenser flow	DP, DT, Power, Cond Leaving — four of six	The whole efficiency and high-head branch. Condenser water-side, refrigerant-side and compressor inefficiency can no longer be separated.	NO_DIAGNOSIS + data-quality work order
Evaporator flow / DPT	SP	Starved evaporator and valve hunting.	NO_DIAGNOSIS for evaporator-side classes
Chiller power	Power	The symptom that starts the isolation path.	No efficiency fault can be raised
Compressor lead/lag	DP, SP, DT, Comp Amps, Cond Leaving	Residuals shift at every stage change.	Gate fails on transitions
Condenser water entry temp	DP, DT, Power, Cond Leaving	Head behaviour cannot be normalised for water conditions.	NO_DIAGNOSIS for condenser-side classes

6. Data and Instrumentation Requirements

The column on the right is for the site survey. A feature cannot be committed to a site until its required signals are confirmed available and trustworthy.

Signal	Required for	Criticality	Available on site?
Suction pressure	DP, SP, DT, Comp Amps	Essential	
Discharge pressure	DP, DT, Comp Amps	Essential	
Discharge temperature	DT	Essential	
Compressor amps	Comp Amps	Essential	
Chiller power (external meter preferred)	Power — the start of the isolation path	Essential	
Evaporator entering water temperature	SP; evaporator ΔT	Essential	
Evaporator leaving water temperature	SP, Power; evaporator ΔT	Essential	
Condenser entering water temperature	DP, DT, Power, Cond Leaving; condenser ΔT	Essential	
Condenser leaving water temperature	Cond Leaving; condenser ΔT	Essential	
Condenser flow	DP, DT, Power, Cond Leaving; efficiency proxy	Highest leverage single measurement	
Evaporator flow / DPT	SP; efficiency proxy	High	
Compressor lead/lag state	Five of six models	High	
Wet bulb temperature	Cooling tower assessment	Medium	
Outside air temperature	Air-cooled equivalents	Medium	

7. What Is ML, What Is Rules, What Is the LLM

Question	Decided by	Never decided by
Is this reading abnormal, and by how much?	The ML model, as a residual	Rules or the LLM
Is the equipment fit to be judged at all?	A deterministic gate	ML or the LLM
Has the pattern lasted long enough?	A deterministic persistence rule	The LLM
Which fault class is this?	The deterministic isolation path over the residuals	The LLM
How confident should we be?	Gate results plus the rule path	The LLM's tone
What does it mean in plain English?	The LLM	—
What should we do about it?	LLM proposes, rules rank, a human decides	The LLM alone
What priority does the work get?	A deterministic formula	The LLM
Is this person allowed to do this?	Plain software — the Control Plane	ML or the LLM
Did the repair actually work?	Post-work residuals judged by a deterministic rule	The closure note or the LLM

8. Open Questions for SME Sign-Off

These are the decisions that block build work. Each needs a value, a range or a yes/no from the SME. Proposed values are starting points for discussion, not commitments.

ID	Question	Why it blocks	Proposed starting point	Decision
Q1	Is condenser flow measured at the target sites, and is the signal trustworthy?	Four of six models depend on it. Without it, most of the fault classes cannot be separated.	Meter it; otherwise scope the site to evaporator-side and sensor-fault classes only	
Q2	How is evaporator flow derived — measured, from DPT, or assumed constant?	Determines whether rSP is trustworthy and whether the efficiency proxy is meaningful.	Confirm provenance per site before any rSP-based class is enabled	
Q3	What is the minimum load for a valid diagnosis?	Below it the models are outside their trained envelope.	Confirm per machine type	
Q4	How long after a start or stage change before residuals are trusted?	Transients produce false abnormalities.	To be set with SME	
Q5	How long after a leaving-water setpoint change?	Same reason.	To be set with SME	
Q6	Persistence window — one value or per fault class?	Too short creates noise; too long delays real faults.	20–30 minutes as a common default	
Q7	Healthy chilled-water ΔT band per machine.	Drives the ΔT health check and a common early warning.	Design ΔT from the machine data sheet; sustained deviation flagged	
Q8	Condenser approach temperature thresholds for fouling.	Separates fouling severity levels.	Per OEM data sheet, confirmed with SME	
Q9	Time constants that separate low condenser flow from fouling.	Both raise rDP and rCWL; only the shape differs.	Low flow = spiky/intermittent; fouling = steady drift over weeks	
Q10	Refrigerant-side high head — split overcharge from non-condensables, or keep combined?	Affects the corrective action given to the technician.	Keep combined until instrumentation supports a split	
Q11	Starved evaporator — confirm undercharge and restriction stay combined.	The available sensors cannot separate them.	Keep combined and label the ambiguity honestly	
Q12	Volatility threshold that defines expansion valve hunting.	Distinguishes hunting from steady starvation.	To be set from observed data with SME	
Q13	Sensor drift thresholds per sensor type before declaring sensor bias.	Stops instrument faults being reported as equipment faults.	Per sensor class, confirmed with SME	
Q14	Is compressor lead/lag state reliably available from the BMS?	Five of six models use it.	Confirm per site during survey	
Q15	What proves a condenser cleaning worked?	Defines the verification PASS criteria.	rPwr, rDP and rCWL back inside band, held several days across a representative load range	
Q16	Re-baseline policy after an overhaul — who authorises, and how long must the machine run first?	A repaired machine has a new normal; baselining too early locks in a transient.	SME authorises; minimum steady running period to be agreed	
Q17	Is the cooling tower in scope for phase one?	Tower condition drives condenser water behaviour and can mimic chiller faults.	Recommend in scope for assessment, out of scope for control	
Q18	Relative weighting of criticality, SLA and production impact in the priority formula.	Determines what crews do first.	Draft weighting to be reviewed with operations	
Q19	Which actions require SME sign-off rather than supervisor approval?	Sets the approval matrix for refrigerant-circuit work.	Refrigerant-circuit work proposed as SME sign-off	
Q20	Are there fault classes missing from section 5.3?	The register is the committed scope.	Open — SME to add	

9. Decision Log

Filled in during and after the review session. Anything recorded here is what gets built — the main document is updated from this log.

ID	Mark	Decision agreed	Owner	Date	Reflected in main doc?